

Group Project Deliverables



Merve Tekgürler

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Expectations



The group project requires you to work in small groups (3-4 students) and develop a project that answers a clearly defined question using computational methods.

- 01** Showcase your coding skills, applied to a new question and context
- 02** Develop an argument and support it with data
- 03** Make your work intelligible to your classmates
- 04** Practice what coding can do in Humanities and Humanistic Social Sciences research

What you DON'T need to do

Remember, this is a class project for an introduction course. The learning objectives are for you to **develop a question, find a dataset, analyze it, and answer that question**. Completing a project from start to finish is an important skill!



01 DON'T feel pressured to say something brand new

It is ok if what you come up with is not brand new or ground breaking. What matters is that you get to practice making arguments with DH methods.

02 This is NOT your only opportunity to do DH

This might be the first time you are working with this dataset/approach/question but this does not have to be the last. You can always revisit your work later.

03 You DON'T need to cover everything

There are many possible directions in any project. You don't need to pursue all leads. Stick with the ones that make sense for your argument.

Main GOAL

This is a group project. The main goal is to build an argument, *together*. Collaboration is an essential part of digital humanities.

Deliverable One

SHORT PAPER

WHAT IT'S ABOUT

You will write a short paper (3-4 pages) summarizing your findings. This paper should clearly identify the question you studied and your approach to it. You must describe the data and the methods you used to answer this question. What is your answer to the question? How do you support it with data?

WHAT TO INCLUDE

Your paper must include at least one data visualization. Address the limitations of your approach. You should highlight where your methods were not successful or where the data was lacking. What are some potential future research directions? You do not need to include your code in the paper but you will need to submit and present your code separately.



WHAT TO PAY ATTENTION TO

This is a collaborative paper, which means your group must develop a shared voice. While you may divide the writing of individual sections among team members, everyone is collectively responsible for the quality and coherence of the final product. Read and revise the paper together, ensuring it tells a unified and compelling story.

Required Sections

Research Questions: A statement of the project's core research questions.

Data: A description of the dataset that the project used for the analyses.

Methods and Approaches: A description of the methods that you are using, and a description of the approach that is the focus of your investigation.

General Reasoning or Discussion: An explanation of how the data and approach come together to help answer or evaluate your core research questions.

Results: What were the results of your analysis? Include visualizations.



Deliverable Two

JUPYTER NOTEBOOK



WHAT IT'S ABOUT

You are required to give a 10-minute presentation of your group project. A functioning Jupyter Notebook is the only acceptable format. You will create one notebook as the entire group. Walk us through the notebook, your findings, and the choices you made in the research process.

WHAT TO INCLUDE

Your notebook should have three sections. The first section will introduce us to your dataset and your question. Show us what is in the data with some examples and explanations. The second section will contain your methods. Tell us what your approaches were and why they make sense for your question. The third section will give us the answer to your question. Visualize your data and report your results.

WHAT TO PAY ATTENTION TO

Your code should be clear and understandable. You will run it live on Zoom, so make sure that it works ahead of time. Comment your code or explain in markdown cells what the code is doing. If you are using code from another researcher, make sure to reference them. Same goes for generative AI. You can use any resources you need but be prepared to explain what the code is doing and why.

Example Jupyter Notebook

01 Introduction

From Primary Source to Data

This Jupyter Notebook is a step-by-step tutorial of the text extraction and dataset building process for the primary source: The Letters of Lady Montagu, originally published in 1763.

What is the question that your research project is tackling? → "What/why/how did X happen?"

02 Dataset

```
import pandas as pd

# Load the CSV file into a DataFrame
montagu_unedited = pd.read_csv('../Data/montagu/montagu_letters_unedited.csv')
montagu_unedited.head()
```

	filename	title	addressee	location	date	
0	letter_1.txt	LETTER I.	TO THE COUNTESS OF ———.	Rotterdam	Aug. 3. O. S. 1716.	I FL/
1	letter_2.txt	LET. II	TO MRS. S——.	Hague	Aug. 5. O. S. 1716.	I M/
2	letter_3.txt	LET. III	TO MRS. S. C.	Nimeguen	Aug.13. O. S. 1716.	I AM

What does your dataset look like? What does it contain? Where is it from? Show us some examples from the dataset, such as printing what is in the first row and explaining all the columns. This will be different for each group depending on the kind of data they are working with.

Example Jupyter Notebook

03 Methods and Approaches

```
from nltk import pos_tag
nltk.download('averaged_perceptron_tagger_eng')
from nltk.tokenize import word_tokenize

# Sample sentence for POS tagging
sample_sentence = sentences[0] # First sentence for demonstration

# Apply POS tagging
pos_tags = pos_tag(word_tokenize(sample_sentence))

print(f"POS Tags for the sample sentence: {pos_tags}")
```

Python

What is the method that you are using? How are you applying this method to your data? Why did you select this method? What library/coding approach/model are you using.

Example Jupyter Notebook

04 Results and Visualizations

```
# Let's see the same in a word cloud
# Generate a word cloud from the cleaned frequency distribution
wordcloud_cleaned = WordCloud(width=800, height=400, background_color='white').generate_from_frequencies(dict(top_30))
# Display the cleaned word cloud
plt.figure(figsize=(12, 6))
plt.imshow(wordcloud_cleaned, interpolation='bilinear')
plt.axis('off') # Hide the axes
plt.title("Top 30 Words in Montagu Letters After Removing Words Common in More Than 50 percent of Letters")
plt.show()
```

Python

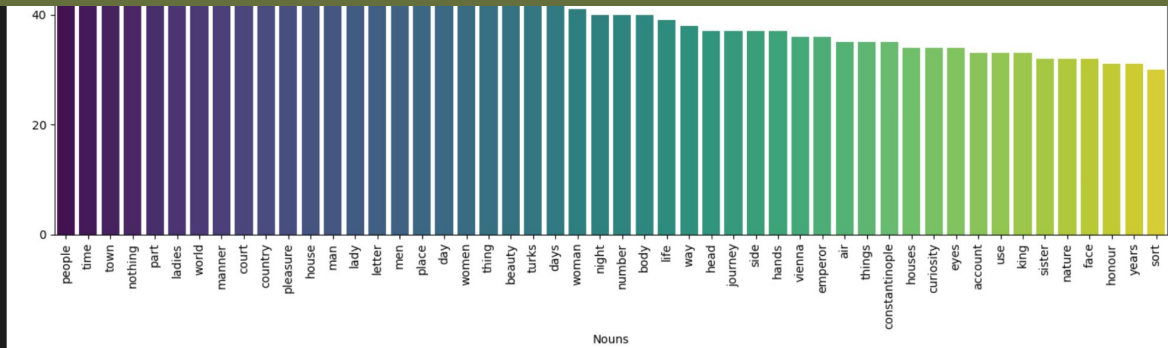
Top 30 Words in Montagu Letters After Removing Words Common in More Than 50 percent of Letters



Visualize your findings. If you have numerical results, report them in a table. Show us the code for visualizations as well as the visualizations themselves

Example Jupyter Notebook

05 Conclusion



Nouns already tell a different story about our data. In many ways, they get us closer to the contents of these texts

Answer your question with a concrete argument → "We argue that ..."